

WASP-12b

R-0295

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-12_240215 · created 2026-07-17T19:25:16+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2460355.7083 +/- 0.001 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.149 +/- 0.015
Transit depth (Rp/Rs)^2	2.23 +/- 0.45 [%]
Orbital Inclination (inc)	86.54 +/- 1.82
Airmass coefficient 1 (a1)	1.407 +/- 0.053
Airmass coefficient 2 (a2)	-0.299 +/- 0.013
Scatter in the residuals of the lightcurve fit is	3.64 %
Best Comparison Star	None
Optimal Aperture	3.99
Optimal Annulus	6.64
Transit Duration (day)	0.1328 +/- 0.0026

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.149 ± 0.015 vs 0.1178 (deviation 0.97σ)
Duration fit vs published	0.1328 d vs 0.125 d (deviation 1.4σ)
Mid-time O–C	-3.5 min
Depth SNR	4.6
Residual scatter	3.64 %

Light curve

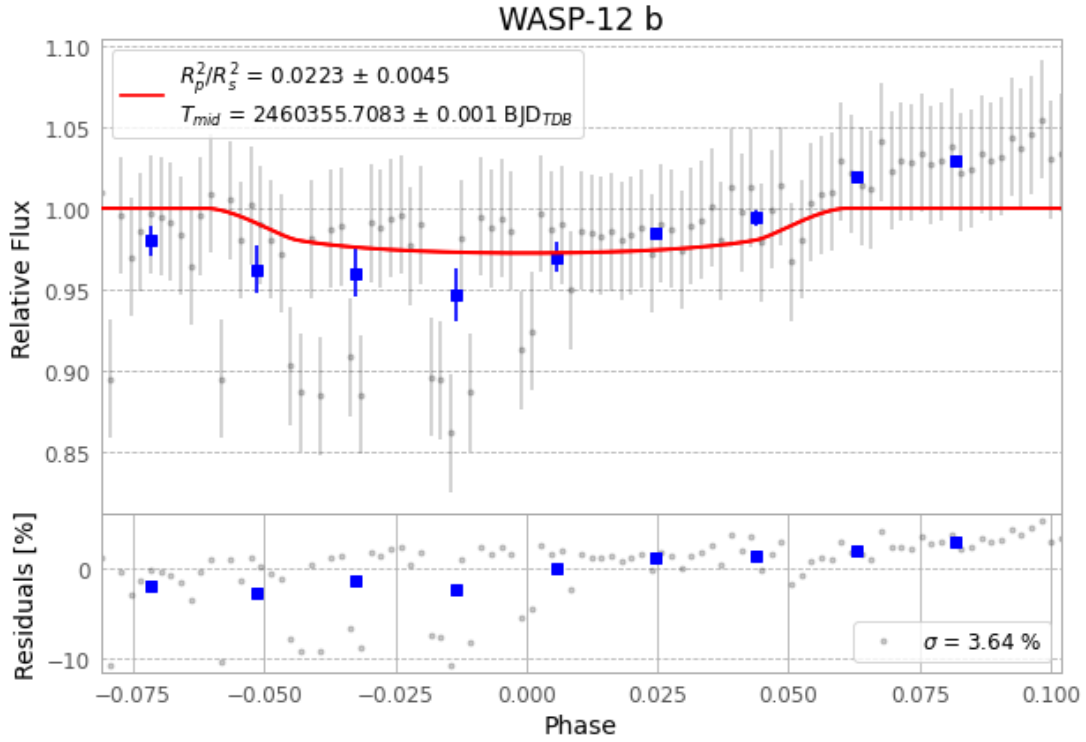


Figure 1 — FinalLightCurve_WASP-12 b_2024-02-14.png (SHA-256 0e1b16e79bb03c17...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [367, 244], 1 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 278efc00d6fd7a40...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit f1e3fa1

Data provenance

97 FITS frames (64 MB), WASP-12240215024515.FITS → WASP-12240215073315.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-12-240215.log (a126d9e67650...), FinalLightCurve_WASP-12 b_2024-02-14.png (0e1b16e79bb0...), FinalLightCurve_WASP-12 b_2024-02-14.csv (fbffbc597292...)

Compute resources

Wall time 115.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-17 19:25Z.